TrustVault

API-Based Credential Verification Network

Software Requirements Specification (SRS)

Version 1.0 | 2025

"Upload once. Verify everywhere. Instantly."

1. Introduction

1.1 Purpose

This document defines the complete Software Requirements Specification (SRS) for TrustVault — an API-based credential verification network. It describes the system's purpose, architecture, website structure, panels, pages, and all functional and non-functional requirements. It is intended for developers, evaluators, and project stakeholders.

1.2 Project Overview

TrustVault solves a core problem: a person's verified data (education, employment, financial, medical) lives in isolated silos. Every time someone applies for a job, a loan, or college admission, they re-upload the same documents, which are then manually verified. This is slow, fraud-prone, and frustrating.

TrustVault replaces the document-upload model with a real-time API verification model:

- Institutions (colleges, banks, hospitals, companies) issue verified credentials directly into the system
- Users hold these credentials in a secure digital Credential Wallet
- Organizations (employers, lenders) query the API to verify instantly — no uploads, no manual checks
- A blockchain trust layer ensures all credentials are tamper-proof

1.3 Scope

- Institution Portal — for colleges, banks, hospitals, companies to issue credentials
- User Credential Wallet — digital profile storing all verified data
- Verifier Portal — for companies/organizations to request and view verifications
- Admin Dashboard — system management and oversight
- Public Website — landing page and documentation
- Blockchain Layer — credential hashing and tamper-proof storage
- API Layer — the core engine connecting all parties

1.4 Definitions

Term	Definition
Credential	A verified piece of data issued by an institution (e.g., degree, salary slip)
Wallet	User's digital profile storing all their verified credentials
Verifier	Any organization requesting credential verification (employer, bank, etc.)
Institution	Entity that issues verified credentials (university, hospital, bank)

Term	Definition
API	Application Programming Interface — the connection layer between parties
Blockchain	Distributed ledger that stores credential hashes for tamper-proof verification
Hash	A unique fingerprint of data — changes if the data is altered
KYC	Know Your Customer — identity verification process

2. Problem Statement & Solution

2.1 Current System Problems

#	Problem	Impact
1	Repeated document uploads	Users upload the same documents 10+ times across platforms
2	Manual verification	Takes days/weeks; error-prone and inconsistent
3	Fraud risk	Fake or edited PDFs can be submitted with no automated detection
4	No interoperability	College, bank, hospital, employer systems don't talk to each other
5	Poor UX	Time-consuming and frustrating for both users and organizations

2.2 Our Solution — At a Glance

TrustVault introduces a Universal Credential Network built on three pillars:

Pillar	What It Does
Institution-Issued Credentials	Data comes directly from verified sources, not users — eliminating fraud at the root
API Verification Layer	Organizations verify credentials in real-time with a single API call — no manual checking
Blockchain Trust Layer	Credential hashes are stored on-chain, making tampering detectable and impossible to hide

2.3 Core Value Proposition

❌Old Way: 'Please upload your documents' → ✅New Way: 'Verified in 0.3 seconds'

3. System Architecture

3.1 High-Level Architecture

TrustVault is a multi-stakeholder platform with four primary actors and four system layers:

Layer	Components	Technology
Frontend	Public Website, User Wallet, Institution Portal, Verifier Portal, Admin Panel	React.js, Tailwind CSS
Backend API	REST API Gateway, Auth Service, Verification Engine, Notification Service	Node.js / Express
Database	User data, credentials, audit logs, sessions	PostgreSQL + Redis (cache)
Blockchain	Credential hash storage, tamper verification	Ethereum / Polygon (smart contracts)

3.2 Core API Endpoints

These are the primary API routes that power the entire verification network:

Endpoint	Method	Description
POST /verify-degree	POST	Verify a user's educational credential in real-time
POST /verify-income	POST	Verify income/employment status with user consent
POST /verify-identity	POST	KYC identity verification
POST /verify-medical	POST	Verify medical records with user permission
GET /wallet/{userId}	GET	Retrieve a user's credential wallet (with auth)
POST /issue-credential	POST	Institution issues a new verified credential
POST /consent/grant	POST	User grants permission for a verification
POST /consent/revoke	POST	User revokes permission from a verifier
GET /audit-log/{userId}	GET	View all verification history for a user

4. Website Structure — All Pages & Panels

TrustVault has 5 distinct portals/sections. Here is a complete breakdown of every page and what it does.

4.1 Public Website (trustvault.io)

The public-facing site for awareness, onboarding, and API documentation.

Page 1: Home / Landing Page

- Hero section with tagline: 'Upload once. Verify everywhere. Instantly.'
- Animated flow diagram showing: Institution → Credential → Wallet → Verifier API
- Live stats counter: verifications done, institutions connected, fraud prevented
- 'Get Started' CTA for users, institutions, and verifiers (3 separate flows)
- Testimonials and trust badges

Page 2: How It Works

- Step-by-step visual walkthrough of the full journey
- Three tabs: User Journey / Institution Journey / Verifier Journey
- Interactive demo — click through a mock verification flow

Page 3: For Institutions

- Explains the institution onboarding process
- Lists supported credential types (degree, employment, medical, financial)
- API integration guide and SDKs
- Pricing for institutional accounts

Page 4: For Verifiers (Organizations)

- Explains how companies/banks integrate the verification API
- Sample API request/response code snippets
- ROI calculator — show time/cost saved vs manual verification
- Sign up for verifier account CTA

Page 5: API Documentation

- Full developer docs — all endpoints, parameters, response formats
- Sandbox environment for testing API calls
- SDK downloads (JavaScript, Python, Java)
- Rate limits, authentication, and error codes

Page 6: Contact / Support

- Support ticket form
- FAQ section covering data privacy, consent, integrations
- Links to privacy policy, terms of service, compliance documents

4.2 User Credential Wallet Portal

This is the user-facing application — the heart of TrustVault from a user's perspective.

Page 1: Registration & KYC

- Sign up with name, email, phone, Aadhaar/PAN (Indian ID)
- KYC verification step — government ID check
- Face match / liveness detection (optional advanced KYC)
- Account activation after successful KYC

Page 2: Dashboard (Home)

- Overview cards: Total credentials, Pending verifications, Recent activity
- Quick action buttons: Share credential, View wallet, Manage consent
- Notification feed: 'XYZ Company requested your degree verification'
- Security score and wallet health indicator

Page 3: My Credential Wallet

- Grid view of all verified credentials (cards with institution logo, date, status)
- Each credential card shows: Type, Issued by, Date, Verified badge, Hash ID
- Click any credential to see full details and blockchain proof
- Download credential as a verified PDF (tamper-proof with QR code)
- Filter by category: Education / Employment / Financial / Medical / Identity

Page 4: Consent Management

- List of all organizations that have requested or received your data
- Toggle to grant or revoke permission per organization per credential
- Consent history — see who accessed what and when
- Set expiry on consents: e.g., 'Allow XYZ Corp to verify my degree for 30 days'
- Emergency revoke all — one click to block all active verifiers

Page 5: Verification History / Audit Log

- Full timeline of every verification request made for your credentials
- Details: verifier name, credential accessed, timestamp, API response
- Export audit log as PDF or CSV
- Flag suspicious activity or unauthorized access

Page 6: Profile & Security Settings

- Edit personal details (name, contact, recovery email)
- Change password, enable 2FA (OTP / Authenticator app)

- Linked devices — see all sessions, revoke remote sessions
- Delete account option with data export

4.3 Institution Portal

Used by colleges, banks, hospitals, and companies to issue and manage credentials.

Page 1: Institution Registration & Verification

- Institution sign-up: Name, type (college/bank/hospital/company), registration number
- Admin manually reviews and approves institution before activation
- Upload official documents: registration certificate, government approval
- Digital signing certificate issued to institution after approval

Page 2: Institution Dashboard

- Summary cards: Total credentials issued, Active users, Verifications requested, Pending approvals
- Charts: Credentials issued over time, Most verified credential types
- Recent activity feed

Page 3: Issue New Credential

- Search user by email, phone, or Aadhaar number
- Select credential type from the institution's allowed types
- Fill credential data form (varies by type — degree fields vs salary fields)
- Preview credential before issuing
- On issue: credential is hashed → stored on blockchain → added to user's wallet
- User gets notified: 'XYZ University issued your BSc CS degree'

Page 4: Manage Issued Credentials

- Table of all credentials issued by this institution
- Filter by type, date, status (active / revoked / expired)
- Revoke credential — if an error was made or credential is no longer valid
- Audit trail per credential — who verified it, how many times

Page 5: API Integration

- Generate and manage API keys for the institution
- Integration guide for the institution's own systems (HR system, student portal, etc.)
- Webhook setup — get notified when your credentials are verified
- Test mode / sandbox environment

Page 6: Institution Settings

- Update institution profile and official details
- Manage admin users and set role-based access (Super Admin, Issuer, Viewer)

- Set credential types this institution is authorized to issue

4.4 Verifier Portal (Employer / Bank / Organization)

Used by companies, banks, and any organization that needs to verify user credentials.

Page 1: Verifier Registration

- Company sign-up with business registration details
- Admin approval process before API access is granted
- Define what credential types the verifier is allowed to query

Page 2: Verifier Dashboard

- Cards: Total verifications done, Success rate, Fraud flags, API usage
- Verification volume chart (daily/weekly/monthly)
- Most verified credential types

Page 3: Run a Verification

- Enter user's email or phone to initiate a verification request
- Select which credentials to verify (degree, income, identity, etc.)
- System sends consent request to user
- Once user approves: system calls API → returns verified result in seconds
- Result display: Verified ✓ / Not Verified ✗ / Pending consent

Page 4: Verification Reports

- Full history of all verifications done by this organization
- Per-verification detail: user, credential type, result, timestamp
- Export as PDF or CSV for compliance/HR records
- Fraud flag reports — credentials that failed blockchain hash check

Page 5: API Management

- View and regenerate API keys
- Usage dashboard — API calls per day, quota remaining
- Configure webhooks for async verification callbacks

4.5 Super Admin Panel

Internal panel for TrustVault team to manage the entire ecosystem.

Page 1: Admin Dashboard

- System-wide stats: Total users, institutions, verifiers, credentials, verifications
- Live API health monitor
- Fraud detection alerts

Page 2: User Management

- Search, view, suspend, or delete user accounts
- View any user's audit log (for compliance investigations)
- Handle user data deletion requests (GDPR / DPDP compliance)

Page 3: Institution Management

- Approve or reject institution registration requests
- View all institutions, their credential types, and issuance stats
- Suspend or deactivate institutions for policy violations

Page 4: Verifier Management

- Approve or reject verifier accounts
- Set API rate limits per verifier
- Monitor unusual API activity and flag potential abuse

Page 5: Blockchain Monitor

- View all on-chain hash records
- Verify system health — confirm hash storage is working
- Transaction logs from the smart contract

Page 6: System Settings

- Manage allowed credential types globally
- Configure rate limits, API quotas, and pricing tiers
- System announcements and maintenance mode

5. Functional Requirements

ID	Module	Requirement
FR-01	Auth	Users must verify identity via KYC before accessing the wallet
FR-02	Auth	All portals must support 2FA (OTP/Authenticator)
FR-03	Credential	Institutions must be able to issue, update, and revoke credentials
FR-04	Credential	Every credential must be hashed and stored on the blockchain upon issuance
FR-05	Consent	Users must explicitly grant consent before any credential is verified
FR-06	Consent	Users must be able to revoke consent at any time, in real-time
FR-07	Verification	API must return verification result in under 2 seconds
FR-08	Verification	Blockchain hash must be checked on every verification request
FR-09	Audit	Every API call, consent, and verification must be logged with timestamp
FR-10	Admin	Admins must approve all institutions and verifiers before activation
FR-11	Notification	Users must receive real-time notifications for all consent requests
FR-12	Security	API access must be secured via OAuth 2.0 and API keys

6. Non-Functional Requirements

Category	Requirement
Performance	API verification response time < 2 seconds for 95% of requests
Scalability	System must handle 10,000 concurrent API requests without degradation
Availability	99.9% uptime SLA — less than 8.7 hours downtime per year
Security	End-to-end encryption (TLS 1.3) for all data in transit
Security	All credential data encrypted at rest using AES-256
Privacy	Full compliance with India's DPDP Act and GDPR principles
Auditability	All actions logged and audit trails retained for 7 years
Reliability	Blockchain hash storage must be redundant across multiple nodes
Usability	All user-facing pages must load in under 3 seconds on 4G mobile
Maintainability	Modular codebase with unit test coverage > 80%

7. Technology Stack

Layer	Technology	Purpose
Frontend	React.js + Tailwind CSS	All portals and public website
Backend	Node.js + Express.js	REST API server and business logic
Database	PostgreSQL	Users, credentials, audit logs (primary storage)
Cache	Redis	Session management, API rate limiting, fast lookups
Blockchain	Ethereum / Polygon	Credential hash storage via smart contracts
Smart Contracts	Solidity	On-chain credential hash functions
Auth	JWT + OAuth 2.0	Authentication and API authorization
File Storage	AWS S3 / Cloudflare R2	Encrypted credential backup storage
Notifications	Firebase / Twilio	Push, SMS, and email notifications
Hosting	AWS / GCP	Cloud infrastructure with auto-scaling
CI/CD	GitHub Actions	Automated testing and deployment pipeline
Monitoring	Datadog / Sentry	Performance monitoring and error tracking

8. User Roles & Permissions

Role	Who	Key Permissions
End User	Any individual (student, employee, patient)	View wallet, manage consent, view audit log
Institution Admin	College registrar, HR manager, bank officer	Issue credentials, manage issued credentials, view stats
Institution Issuer	Staff at institution	Issue credentials only (limited role)
Verifier Admin	Employer HR, bank loan officer	Run verifications, view reports, manage API keys
Super Admin	TrustVault team	Full system access — approve institutions, manage all data
API Consumer	Developer integrating the API	Programmatic access via API keys (scoped)

9. End-to-End User Flows

9.1 Credential Issuance Flow

- Step 1: Institution logs into Institution Portal
- Step 2: Searches user by phone/email/ID
- Step 3: Fills credential form (e.g., Degree: BSc CS, Year: 2024, Grade: A)
- Step 4: Submits → Backend hashes the credential data
- Step 5: Hash is written to the blockchain via smart contract
- Step 6: Credential appears in user's Credential Wallet
- Step 7: User receives push notification: 'New credential issued by XYZ University'

9.2 Verification Flow

- Step 1: Verifier (employer) opens their portal and starts a verification
- Step 2: Enters user's email and selects 'Verify Degree'
- Step 3: System sends consent request to user (push/SMS/email)
- Step 4: User sees request on their dashboard and approves
- Step 5: Verifier's API call goes through → Backend retrieves credential
- Step 6: Backend fetches the hash from the blockchain and compares
- Step 7: If hash matches → returns { verified: true } in < 2 seconds
- Step 8: Verifier sees: ✓ VERIFIED — BSc CS, XYZ University, 2024
- Step 9: Audit log updated for both user and verifier

10. Final Summary

"TrustVault: Upload once. Verify everywhere. Instantly."

TrustVault is a complete credential verification ecosystem. It replaces the broken document-upload model with a real-time, institution-sourced, blockchain-secured API verification network. With 5 distinct portals (Public Website, User Wallet, Institution Portal, Verifier Portal, Admin Panel), a robust REST API, and a blockchain trust layer, it solves the five core problems of the current system:

Problem Solved	How
Repeated uploads	Credentials issued once, stored in wallet — never re-uploaded
Manual verification	Instant API response replaces manual document checks
Fraud	Institution-sourced data + blockchain hash verification eliminates fake documents
No interoperability	Common API layer connects education, banking, healthcare, and employment
Poor UX	One-click consent + instant results for both users and verifiers

TrustVault is not just a product — it is the infrastructure for a trusted digital identity ecosystem.